

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME			
CLASS		INDEX NUMBER	

BIOLOGY

9744/02

Paper 2 Structured Questions

29 August 2017

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	12
2	14
3	10
4	15
5	13
6	12
7	12
8	12
Total	100

This document consists of **23** printed pages and **1** blank page.



Answer **all** questions.

- 1 Fig. 1.1 shows a ligand binding to the G protein-coupled receptor (GPCR) which is embedded on the cell surface membrane.

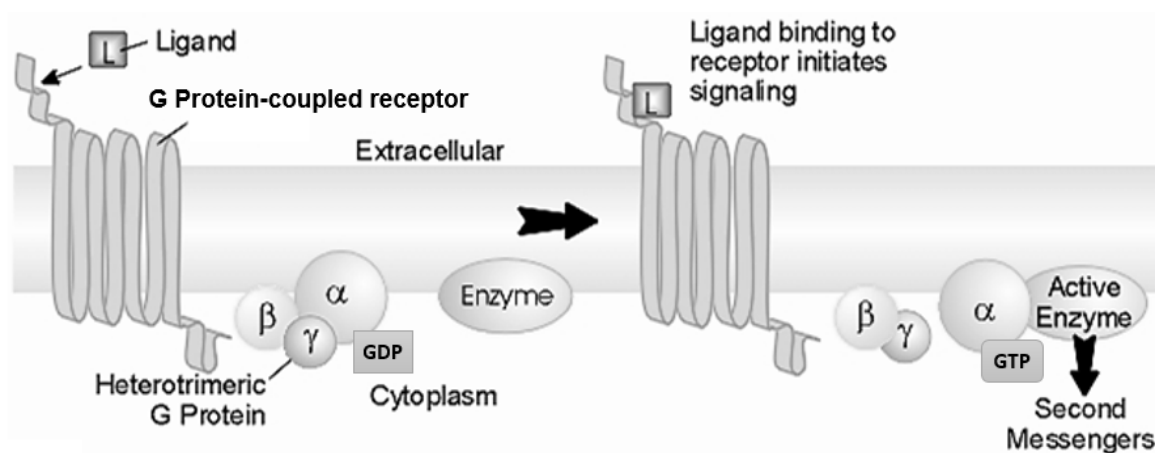


Fig. 1.1

- (a) (i) Identify the ligand in Fig. 1.1.

..... [1]

- (ii) Explain why ligand mentioned in (a)(i) unable to pass through the cell surface membrane.

.....

 [2]

- (b) With reference to Fig. 1.1, describe what happens when the ligand binds to the GPCR.

.....

 [3]

- (c) With a named example, define "second messengers".

.....

.....

.....

..... [2]

- (d) Explain how intracellular signal is terminated when ligand is released from the receptor.

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.....

..... [2]

- (e) One of the side effects of a particular drug includes non-responsiveness of GPCR to ligands.

Suggest how the drug could have caused such non-responsiveness of GPCR to ligands.

.....

.....

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..... [2]

[Total: 12]

- 2 Fig. 2.1 is a photomicrograph of plants cells, with some undergoing mitosis.

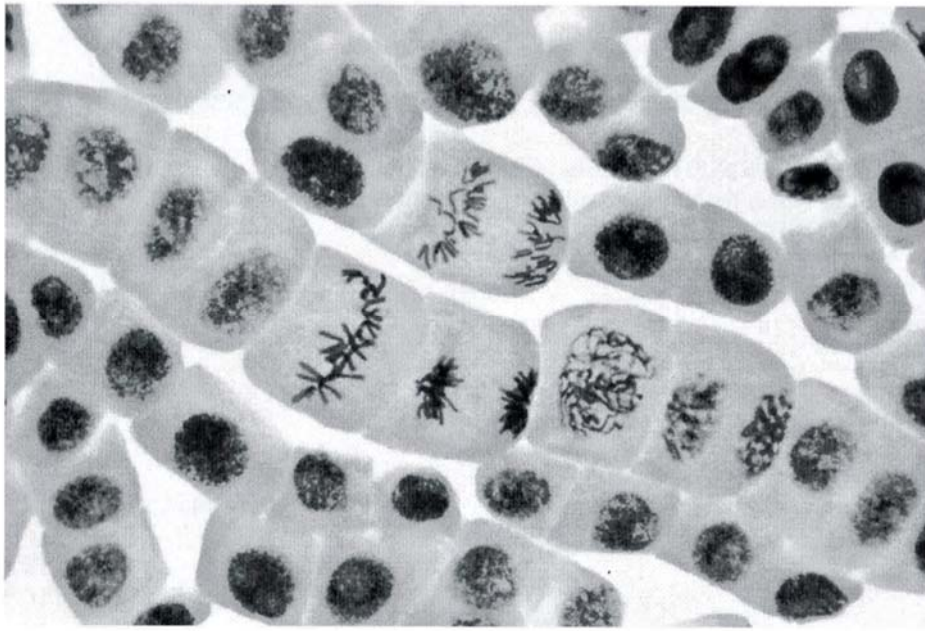


Fig. 2.1

- (a) On Fig. 2.1, use labels and label lines to indicate **one** cell in anaphase stage of mitosis. [1]

- (b) The longest stage of the mitotic cell cycle, interphase, is divided into three phases, **G1, S** and **G2**.

- (i) Describe what happens in the G1 phase.

.....

.....

.....

.....

.....

..... [3]

- (ii) There are various checkpoints in the mitotic cell cycle. One of them is present in the G1 phase called the G1 checkpoint.

Describe the function of G1 checkpoint.

.....

.....

.....

..... [2]

- With reference to two named genes, outline the development of cancer.

[4]

- (i) State the potency level of adult stem cells and their function in our body.

[2]

- [2]

[Total: 14]

- 3** Pure breeding sweet pea plants with purple flowers and long pollen grains were crossed with pure breeding plants with red flowers and round pollen. All the F_1 plants had purple flowers and long pollen grains. These F_1 plants were then allowed to self-pollinate and the seeds produced were grown.

The following results were obtained in this F_2 generation.

4831 purple flowers and long pollen grains
390 purple flowers and round pollen grains
393 red flowers and long pollen grains
1338 red flowers and round pollen grains

- (a)** Explain what is meant by the term “pure breeding”.

.....
..... [1]

- (b)** State the expected phenotypic ratio of the F_2 generation.

..... [1]

- (c)** Using suitable symbols, draw a genetic diagram to explain the observed results of the F_2 generation.

[5]

- (d) Suggest how similar crossing experiments with many different pairs of characters could be used to map the position of genes on the chromosomes of sweet pea plants.

----- [3]

[Total: 10]

- 4 Fig. 4.1 shows the life-cycle of *Aedes aegypti*, which are often vectors of viral diseases like dengue fever, chikungunya and yellow fever.

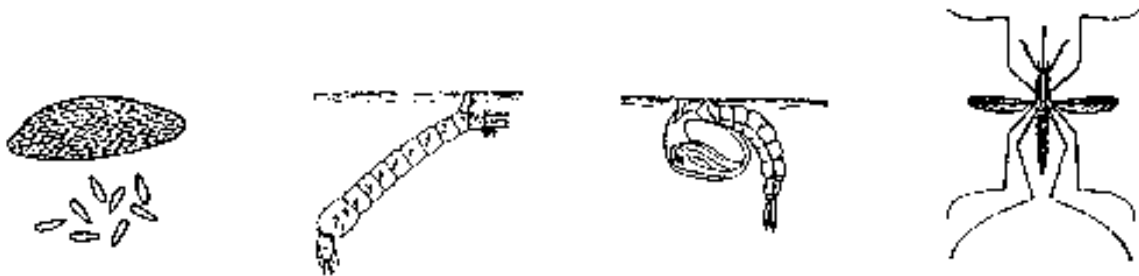


Fig. 4.1

- (a) With reference to Fig. 4.1, name the four stages in a *Aedes aegypti* life-cycle.

[2]

- (b) The following is an extract from an article “Record 2,441 dengue cases reported in Singapore for January” published on Singapore’s The Straits Time website on 2nd Feb 2016.

“SINGAPORE - A total of 636 dengue cases were reported for the week of Jan 24 to 30 - the same number as the previous week - according to the latest figures released by the National Environment Agency (NEA) on Tuesday (Feb 2).

This brings the total number of cases for the first four weeks of the year to 2,441, an unusually high number for January given that it is traditionally the low season for dengue.”

Fig. 4.2 shows the weekly number of dengue cases in Singapore from 2013 to 2016

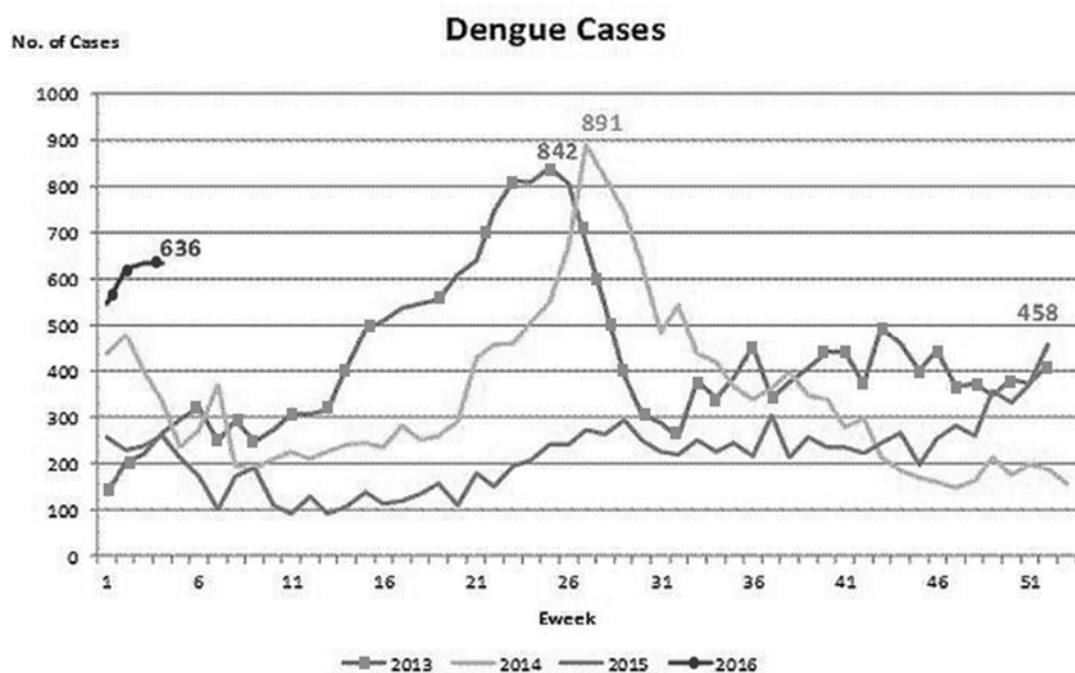


Fig. 4.2

- (i) With reference to Fig. 4.2, describe the general trend in dengue cases in 2013.

.....

.....

.....

..... [2]

Fig. 4.3 shows average monthly temperature in Singapore in Year 2013.

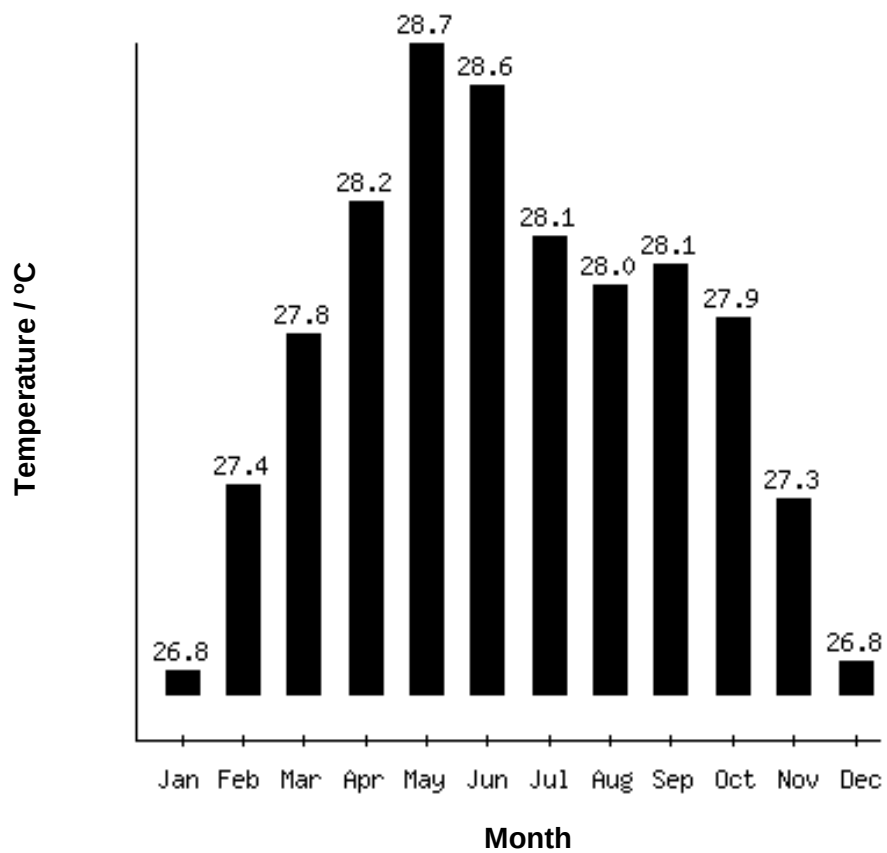


Fig. 4.3

- (ii) With reference to Fig. 4.3, describe the temperature trend in Singapore in 2013.

.....

.....

.....

..... [2]

- (iii) With reference to both Fig. 4.2 and 4.3, account for the relationship between temperature and dengue cases in Singapore in 2013.

.....

.....

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.....

..... [3]

- (c) Fig. 4.4 is a diagram of a dengue virus.

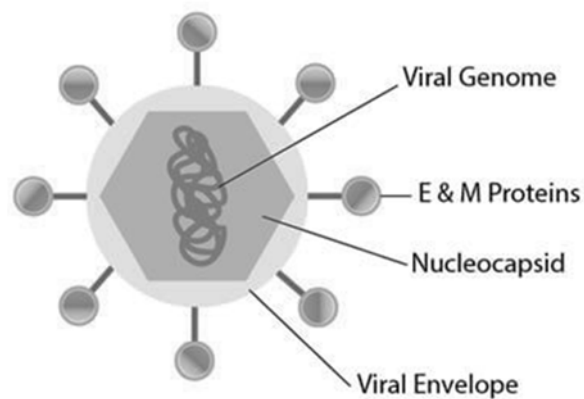


Fig. 4.4

- (i) Describe the viral genome of dengue virus.

.....

..... [1]

- (ii) The dengue virus and influenza virus are quite similar in terms of their structure and reproductive cycle.

Describe one structural similarity between dengue virus and influenza virus.

.....

..... [1]

- (iii) Compare the reproductive cycles of dengue virus and influenza virus.

Giving one difference and two similarities.

Difference:

.....

Similarities:

.....

.....

..... [3]

- (iv) Currently, there are no specific antiviral drugs for the treatment of dengue fever, due to the prevalence of drug resistance in dengue viruses.

Suggest one reason how drug resistance can arise in dengue viruses.

.....

..... [1]

[Total: 15]

5 Fig. 5.1 and 5.2 are diagrams showing transcription and translation.

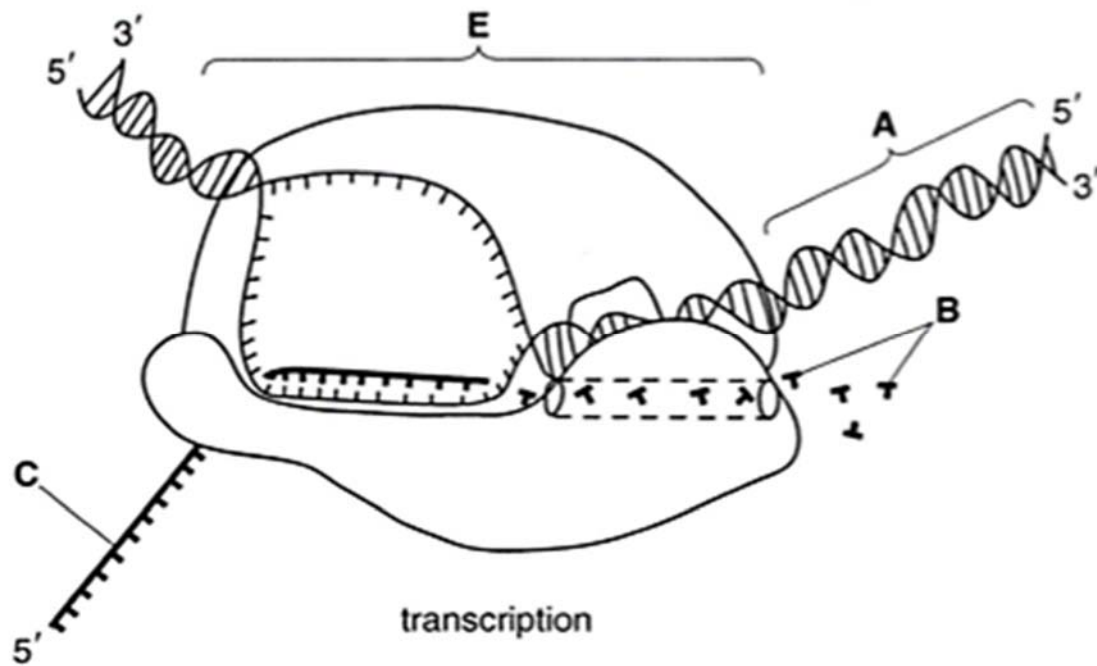


Fig. 5.1

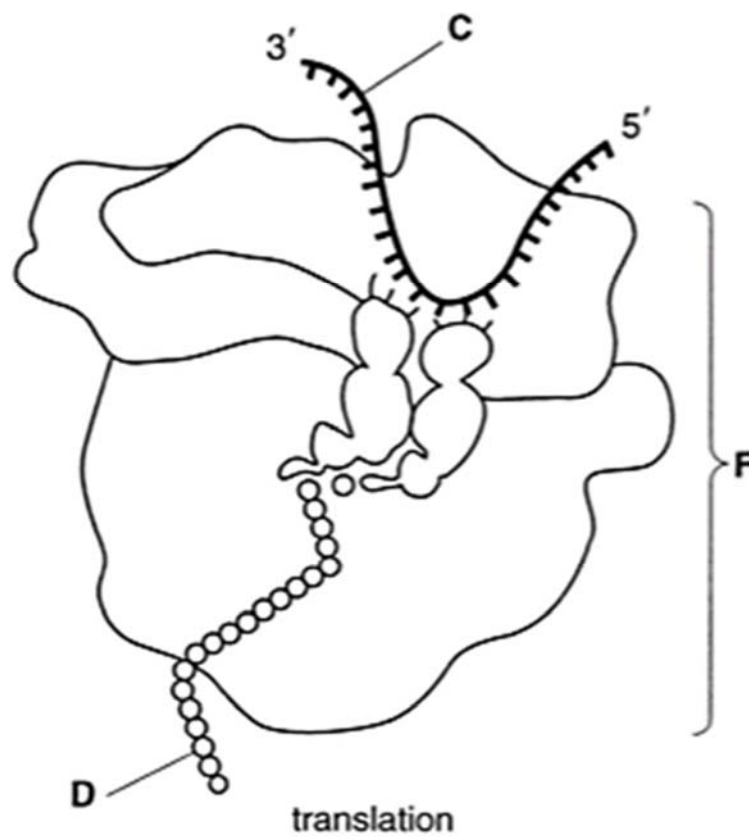


Fig. 5.2

(a) Identify the structures **A** to **C**.

A:

B:

C:

[3]

(b) Describe what happens to **D** after it is being synthesized to form a functional product.

[2]

(c) Describe three ways in which the process of transcription differs from translation.

[3]

(d) Briefly describe how the structure of **F** differs from the structure **E**.

[2]

(e) Fig. 5.3 shows a proteasome degrading a protein.

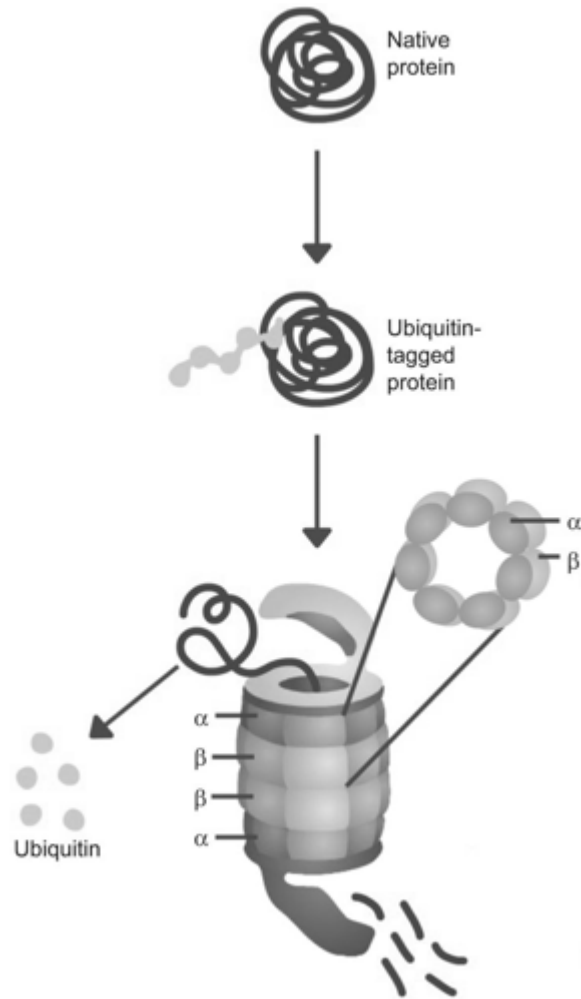


Fig. 5.3

With reference to Fig. 5.3, describe what happens during proteasomal degradation of proteins.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 13]

- 6 The marine threespine sticklebacks, *Gasterosteus aculeatus* is a freshwater fish living in the lakes of British Columbia, Canada as shown in Fig. 6.1.

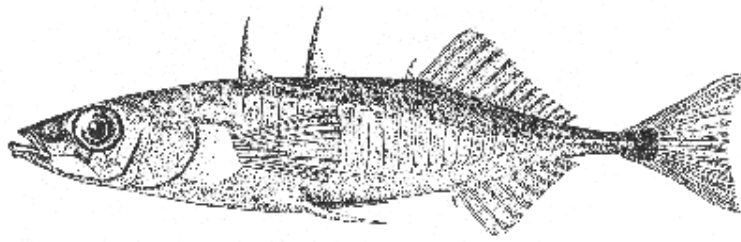


Fig. 6.1

In order to investigate the process of speciation in these populations, three small lakes were studied. Each lake contained two varieties of stickleback: a large, bottom-dwelling variety that fed on invertebrates near the shore and a small, plankton-eating variety that lived in the open water. The probability of breeding between pairs of individuals was measured under laboratory conditions in the following breeding combinations:

I	different varieties from the same lake
II	different varieties from different lakes
III	same variety from different lakes
IV	same variety from the same lake

The data are summarized in Fig. 6.2 below.

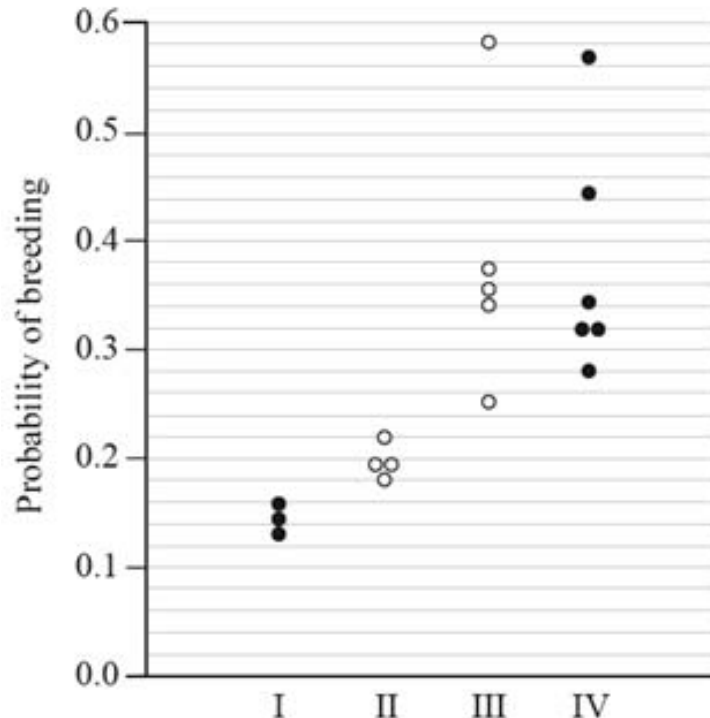


Fig. 6.2

(a) With reference to Fig. 6.2,

- (i)** identify the highest and lowest probabilities of breeding for individuals of the same variety;

..... [1]

- (ii)** describe the differences in probability of breeding between individuals from different lake;

..... [2]

- (iii)** describe the evidence that speciation is taking place in these populations and explain the type of speciation;

..... [3]

- (iv)** explain why all the individuals are still considered the same species.

..... [2]

- Explain why these parasites enable speciation of the marine threespine sticklebacks to occur.

[4]

[Turn over

7 Fig. 7.1 shows how innate immune system protects the body against pathogens.

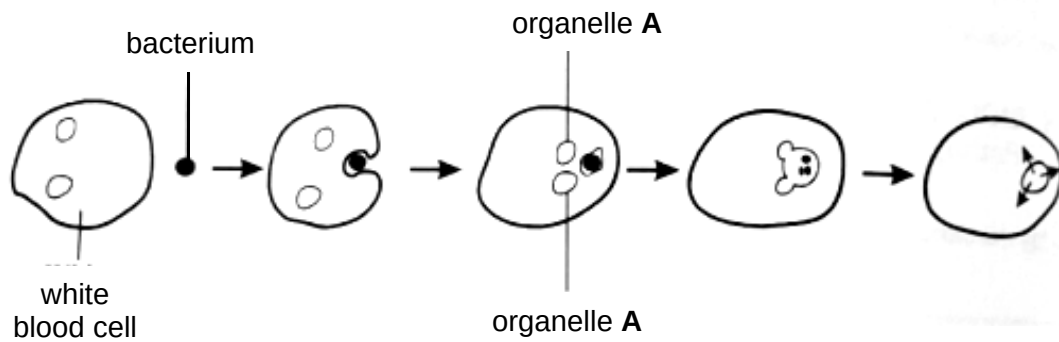


Fig. 7.1

(a) With reference to Fig 7.1,

(i) state the name of the white blood cell and organelle A.

.....

..... [1]

(ii) describe the role of organelle A in the defence against pathogen.

.....

.....

.....

..... [2]

It is found that ingested *Mycobacterium tuberculosis* (*M. tuberculosis*) is able to survive within the macrophage and cause tuberculosis in humans.

Bacillus Calmette–Guérin (BCG) is a vaccine primarily used against tuberculosis. It consists of live attenuated bacteria. In countries where tuberculosis is common, one dose is recommended in healthy babies as close to the time of birth. Babies with HIV/AIDS should not be vaccinated.

- (b) (i)** Explain how BCG vaccination provides long term immunity against tuberculosis.

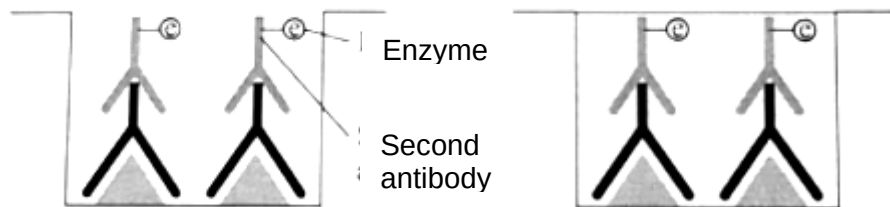
[4]

- (ii)** Suggest why babies with HIV/AIDS should not be vaccinated.

[1]

The diagram illustrates the process of antigen-antibody complex formation in a well of a test dish. On the left, a 'Well in test dish' contains two brown triangles representing 'Antigen'. On the right, the same well contains two Y-shaped black structures representing 'Antibodies' bound to the 'Antigen'.

Step 2: Sample of patient's blood is added to the well. If antibodies are present, they bind to the antigen



Step 4: The well is washed. A solution is added which changes colour if the enzyme is present. A colour change shows that the person has antibodies against *M. tuberculosis*

(c) Predict and explain if the color of the solution will change if the patient is infected with influenza virus.

[4]

[Turn over

- 8** Corals are simple marine animals and usually exist in colonies of thousands of individuals. Zooxanthellae are group of unicellular algae that can photosynthesize. They live within cells of the coral and have a symbiotic relationship.

Corals absorb calcium carbonate from the sea to build their skeletons which provides structural support. Coral reefs provide home for about 25% of known fish species.

Corals are sometimes mistaken for members of plant kingdom.

- (a)** State one way in which coral cells differ from plant cells

[1]

Coral reefs are at risk of damage by human activities. A study was conducted to see the effects of climate change on coral reefs.

Coral reef sites were subjected to two different environmental conditions i.e. exposed site and sheltered site. Coral reefs in exposed site was exposed to climate change environmental conditions. Coral reefs in the sheltered site was exposed to normal environmental conditions.

Table 8 shows coral cover area at exposed and sheltered sites.

Table 8

Experimental site		Area of healthy coral reef/m ²	Average area of healthy coral reef/m ²
Exposed Site	Site 1	120	
	Site 2	100	
	Site 3	150	
Sheltered Site	Site 1	82	
	Site 2	75	
	Site 3	69	

- (b)** With reference to Table 8,

- (i)** complete Table 8 by calculating the average area of healthy coral reef in exposed and sheltered site. Show your working below.

[1]

- (ii) conduct a t -test on the given data and determine if the difference in mean area of healthy coral reef in exposed and sheltered site are statistically significant.

standard deviation $s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$

t -test $t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$ $v = n_1 + n_2 - 2$

Key to symbols

s^* = standard deviation

$\sum$ = 'sum of'

$\bar{x}$ = mean

n = sample size (number of observations) x = observation

v = degrees of freedom

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01
df									
1	0.000	1.000	1.378	1.963	3.078	6.314	12.71	31.82	63.66
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169

[4]

[6]

[6]

[Total: 12]

